

# SAFETY DATA SHEET (GHS)

## 1,4-Di-isopropylbenzene (DIPB) | CAS: 100-18-5 | Formula: C<sub>12</sub>H<sub>18</sub>

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

### DATA SOURCES CONSULTED

- Sigma-Aldrich Safety Data Sheet — 1,4-Diisopropylbenzene, product Aldrich-126276, <https://www.sigmaaldrich.com/>
- ChemicalBook — 1,4-Diisopropylbenzene, CAS 100-18-5, <https://www.chemicalbook.com/>
- ILO/NIOSH International Chemical Safety Card (ICSC) 1714 — Diisopropylbenzene (mixture)
- Accessed 2026-06-13

### SECTION 1 — IDENTIFICATION

**Product identifier:** 1,4-Di-isopropylbenzene (p-DIPB)

**CAS Number:** 100-18-5

**Chemical formula:** C<sub>12</sub>H<sub>18</sub>

**Molecular weight:** 162.27 g/mol

**Synonyms:** p-Diisopropylbenzene; 1,4-bis(1-methylethyl)benzene

**Process role:** Heavy by-product/poly-alkylation product formed during the alkylation step of cumene production (di-isopropylbenzene isomers form alongside cumene and may be recycled via transalkylation). PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

**Emergency telephone:** [Plant emergency response number — to be inserted by PPCL HSE]

### SECTION 2 — HAZARD IDENTIFICATION

**GHS classification:** Skin Irrit. 2 (H315); classified as a combustible liquid

**Signal word:** Warning

**Hazard statements (H-codes):** H315 (Causes skin irritation)

**Precautionary statements (P-codes):** P264, P280, P302+P352, P332+P313, P362+P364, P501

**GHS pictograms:** GHS07 (Exclamation mark)

**Other hazards:** DATA-QUALITY NOTE: available supplier (Sigma-Aldrich) SDS data for this CAS number lists H315 (Skin Irrit. 2) and a Pensky-Martens closed cup flash point of 80°C, with the product noted as a combustible liquid. H411 (Aquatic Chronic 2, aquatic toxicity) and aspiration hazard (H304) — both plausible for an alkylbenzene of this type, by analogy with cumene (CAS 98-82-8) — were NOT confirmed in the sources reviewed for this SDS. RECOMMEND: verify the full classification (including possible H304/H411) against the current ECHA C&L Inventory and supplier SDS, particularly given DIPB's structural similarity to cumene (both low-viscosity alkylbenzenes with aspiration potential).

### SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

**Substance:** 1,4-Di-isopropylbenzene, CAS 100-18-5, typically >97% purity; may be present as part of a mixed di-isopropylbenzene isomer stream (1,3- and 1,4- isomers, and minor 1,2-) in the actual process — confirm isomer distribution against PPCL process data

### SECTION 4 — FIRST AID MEASURES

**Inhalation:** Remove to fresh air. Obtain medical attention if symptoms persist.

**Skin contact:** Wash with plenty of soap and water. Remove contaminated clothing. Obtain medical attention if irritation persists (H315).

**Eye contact:** Flush with copious water for at least 15 minutes. Obtain medical attention if irritation persists.

**Ingestion:** Rinse mouth. Do not induce vomiting (risk of aspiration into lungs for low-viscosity hydrocarbons — see Section 2 data-quality note). Obtain medical attention immediately.

**Notes to physician:** If aspiration is suspected following ingestion, monitor for chemical pneumonitis (consistent with the precautionary approach for alkylbenzene aspiration hazards)

## SECTION 5 — FIREFIGHTING MEASURES

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**Suitable extinguishing media:** Dry chemical, CO<sub>2</sub>, alcohol-resistant foam, water spray (for cooling)

**Unsuitable extinguishing media:** High-volume water jet on a flowing liquid fire

**Specific hazards:** Combustible liquid — flash point ~80°C (Pensky-Martens closed cup); will burn if heated above flash point and exposed to an ignition source. Vapours may form explosive mixtures with air at elevated temperature.

**Special protective equipment for firefighters:** Self-contained breathing apparatus (SCBA) and full protective clothing

**Hazardous combustion products:** Carbon monoxide, carbon dioxide, smoke

## SECTION 6 — ACCIDENTAL RELEASE MEASURES

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**Personal precautions:** Avoid skin contact (H315) and ingestion; ensure adequate ventilation; wear appropriate PPE (Section 8)

**Environmental precautions:** Avoid release to drains/watercourses pending confirmation of aquatic toxicity classification (Section 2)

**Containment and cleanup:** Contain spill with inert absorbent material; collect into closed, labelled containers for disposal; avoid generating vapour

**Reference to other sections:** See Section 7 (Handling/Storage) and Section 8 (PPE)

## SECTION 7 — HANDLING AND STORAGE

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**Handling precautions:** Avoid prolonged or repeated skin contact (H315); avoid ingestion (aspiration precaution per Section 4); use in well-ventilated areas

**Storage conditions:** Store in tightly closed containers in a cool, dry, well-ventilated area away from heat and open flame

**Temperature limits:** Keep below flash point (~80°C) during storage and handling where practical

**Pressure:** Atmospheric storage

**Segregation requirements:** Segregate from strong oxidisers

**Special requirements:** None beyond general combustible-liquid handling practice

**Specific end-use recommendations:** If DIPB is recycled via transalkylation back to cumene/benzene, ensure handling procedures account for the cumulative aromatic hydrocarbon exposure profile across the alkylation/transalkylation loop

## SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

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**Occupational Exposure Limits (OEL):** No specific ACGIH TLV or OSHA PEL established for 1,4-di-isopropylbenzene; confirm Thailand OEL if applicable

**TWA:** Not established

**STEL:** Not established

**Ceiling:** Not established

**Skin notation:** Not assigned

**IARC/NTP/OSHA carcinogen status:** Not classified as a carcinogen

**Engineering controls:** General ventilation adequate for typical handling; local exhaust for tasks generating vapour/mist

**PPE — Respiratory:** Not normally required for typical handling; organic vapour respirator if vapour concentrations are elevated

**PPE — Gloves:** Chemical-resistant gloves (e.g., nitrile) — important given H315

**PPE — Eyes:** Safety glasses or chemical splash goggles

**PPE — Body:** Standard work clothing; chemical-resistant apron for bulk transfer

**Biological monitoring:** Not applicable

## SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

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**Appearance:** Colourless liquid

**Odour:** Characteristic aromatic hydrocarbon odour

**Odour threshold:** Not established

**pH:** Not applicable

**Melting/Freezing point:** -17°C

**Boiling point (at 1 atm):** 210°C

**Flash point:** 80°C (Pensky-Martens closed cup)

**Evaporation rate:** Much slower than water

**Flammability (solid/gas):** Not applicable (liquid); classified as combustible

**Flammability limits in air:** Not established — estimated to be similar to other C12 alkylbenzenes (LEL roughly ~0.5-1% v/v)

**Vapour pressure (at 20°C):** Low (<0.1 kPa) — low-volatility heavy aromatic

**Vapour density (air = 1):** 5.6 (estimated from molecular weight)

**Relative density (water = 1):** ~0.857 at 20°C

**Solubility (water):** Practically insoluble

**Partition coefficient (log K<sub>ow</sub>):** ~4.6 (estimated — relatively lipophilic, consistent with potential aquatic bioaccumulation flagged in Section 2)

**Auto-ignition temperature:** Not well established — confirm against authoritative source

**Decomposition temperature:** Not established under normal conditions

**Viscosity:** Not established — expected low (heavy aromatic hydrocarbon)

**Explosive properties:** Not classified as explosive

**Oxidising properties:** Not an oxidiser

**Molecular weight:** 162.27 g/mol

## SECTION 10 — STABILITY AND REACTIVITY

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**Reactivity:** Stable under normal conditions of use and storage

**Chemical stability:** Stable under recommended storage conditions

**Possibility of hazardous reactions:** No particular hazardous reactions known under normal process conditions; can undergo further alkylation/dealkylation (transalkylation) under the acidic catalyst conditions used elsewhere in the process — this is the intended recycle chemistry, not a hazard per se

**Conditions to avoid:** Heat, open flame, contact with strong oxidisers

**Incompatible materials:** Strong oxidisers

**Hazardous decomposition products:** Carbon monoxide, carbon dioxide on combustion

## SECTION 11 — TOXICOLOGICAL INFORMATION

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**Routes of exposure:** Skin contact (H315), inhalation, ingestion

**Acute toxicity — Oral LD<sub>50</sub>:** Not well established for this specific isomer; diisopropylbenzene mixtures generally show low acute oral toxicity (>2,000 mg/kg, rat, estimated)

**Acute toxicity — Dermal LD<sub>50</sub>:** Not well established

**Acute toxicity — Inhalation LC<sub>50</sub>:** Not well established

**Skin corrosion/irritation:** Causes skin irritation (H315)

**Serious eye damage/irritation:** Not classified under the primary classification used here; may cause mild eye irritation on contact

**Respiratory/skin sensitisation:** Not classified as a sensitiser

**Germ cell mutagenicity:** Not classified

**Carcinogenicity:** Not classified as a carcinogen

**Reproductive toxicity:** Not classified

**STOT (single exposure):** Not classified under available data; confirm given structural similarity to cumene (which carries STOT SE 3, H335)

**STOT (repeated exposure):** Not classified

**Aspiration hazard:** Not classified under available data — see Section 2 data-quality note recommending verification given structural similarity to cumene (Asp. Tox. 1, H304)

## SECTION 12 — ECOLOGICAL INFORMATION

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**Aquatic toxicity (LC50/EC50):** Not well established from available sources; estimated log Kow ~4.6 suggests potential for aquatic toxicity/bioaccumulation consistent with other heavy alkylbenzenes — confirm classification (possible H411)

**Persistence and degradability:** Not readily biodegradable (typical of heavy alkylbenzenes)

**Bioaccumulative potential:** Potentially bioaccumulative (log Kow ~4.6)

**Mobility in soil:** Low mobility (low water solubility, high log Kow)

**Other adverse effects:** None specifically identified beyond potential aquatic hazard noted above

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## SECTION 13 — DISPOSAL CONSIDERATIONS

**Disposal methods:** Dispose of via licensed hazardous waste incineration or treatment facility in accordance with local regulations; do not discharge to drains

**Contaminated packaging:** Dispose of as hazardous waste or, if uncontaminated/empty, per local regulations after appropriate purging

**Regulatory reference:** Thailand Hazardous Substance Act B.E. 2535 (verify applicability); local industrial waste regulations

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## SECTION 14 — TRANSPORT INFORMATION

**UN Number:** Not regulated as a dangerous good for transport based on available data (flash point >60°C; confirm for the specific stream used on-site)

**Proper shipping name:** Not applicable / Not regulated

**Transport hazard class(es):** Not classified as a transport hazard based on available data

**Packing group:** Not applicable

**Environmental hazards (marine pollutant):** Confirm against potential H411 classification flagged in Section 2

**Special precautions for user:** None specific identified

**Bulk transport (MARPOL/IBC):** Confirm IBC Code pollution category for bulk marine transport if applicable

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## SECTION 15 — REGULATORY INFORMATION

**Safety and health regulations:** See country-specific notes below

**Thailand:** Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

**EU:** Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

**US:** Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

**ASEAN:** Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

**Chemical safety assessment:** Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

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## SECTION 16 — OTHER INFORMATION

**Revision date:** 2026-06-13

**Revision number:** 1.0

**Key changes from previous version:** Initial issue — compiled for HAZOP PSI (CAS 100-18-5). DATA-QUALITY FLAG: possible additional hazard categories (Asp. Tox. 1 H304, Aquatic Chronic 2 H411) by analogy with cumene (CAS 98-82-8) were not confirmed in sources reviewed — flagged for manual verification.

**Data sources used in this SDS:** See 'Data Sources Consulted' block at top of document

**Disclaimer:** This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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